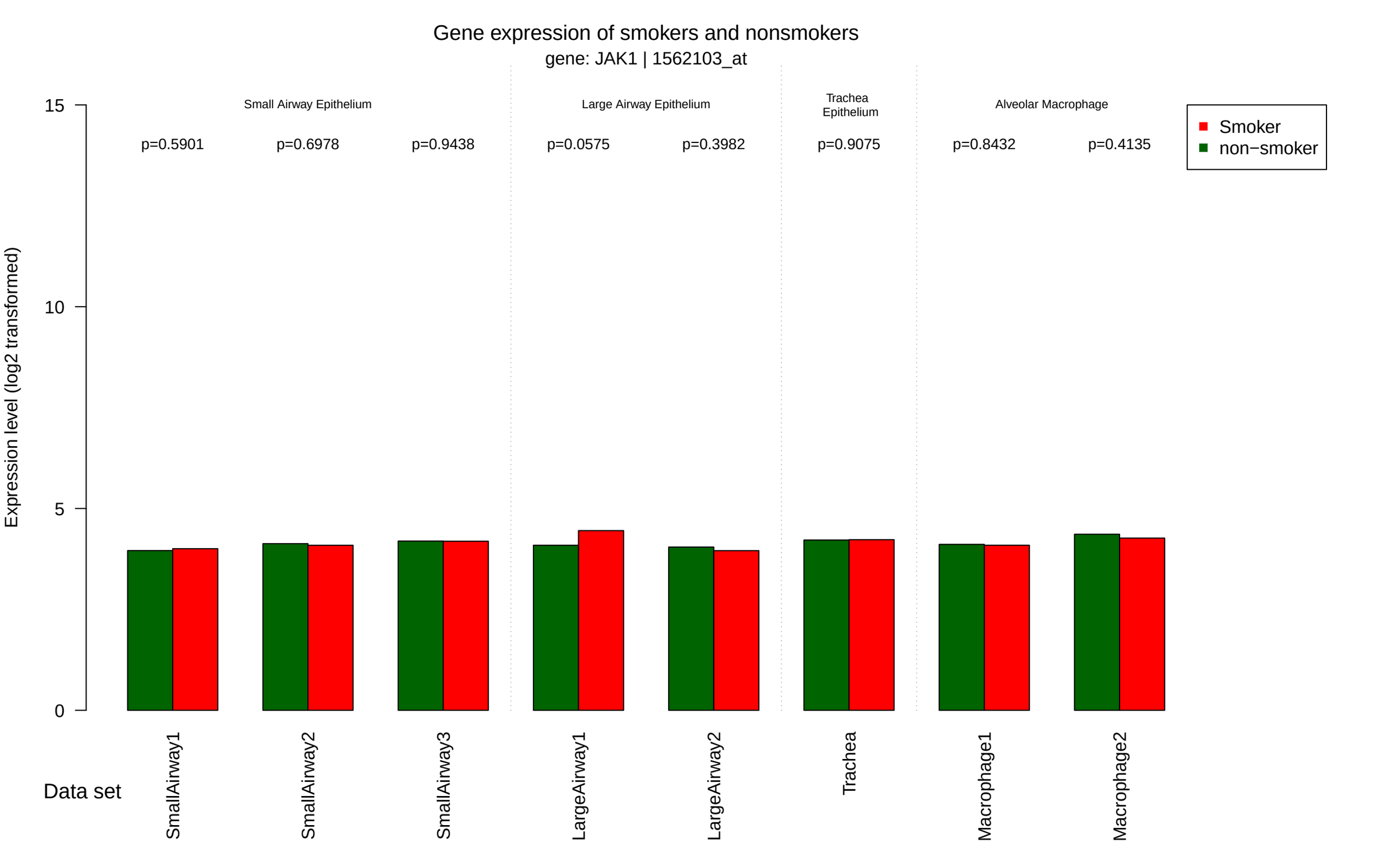
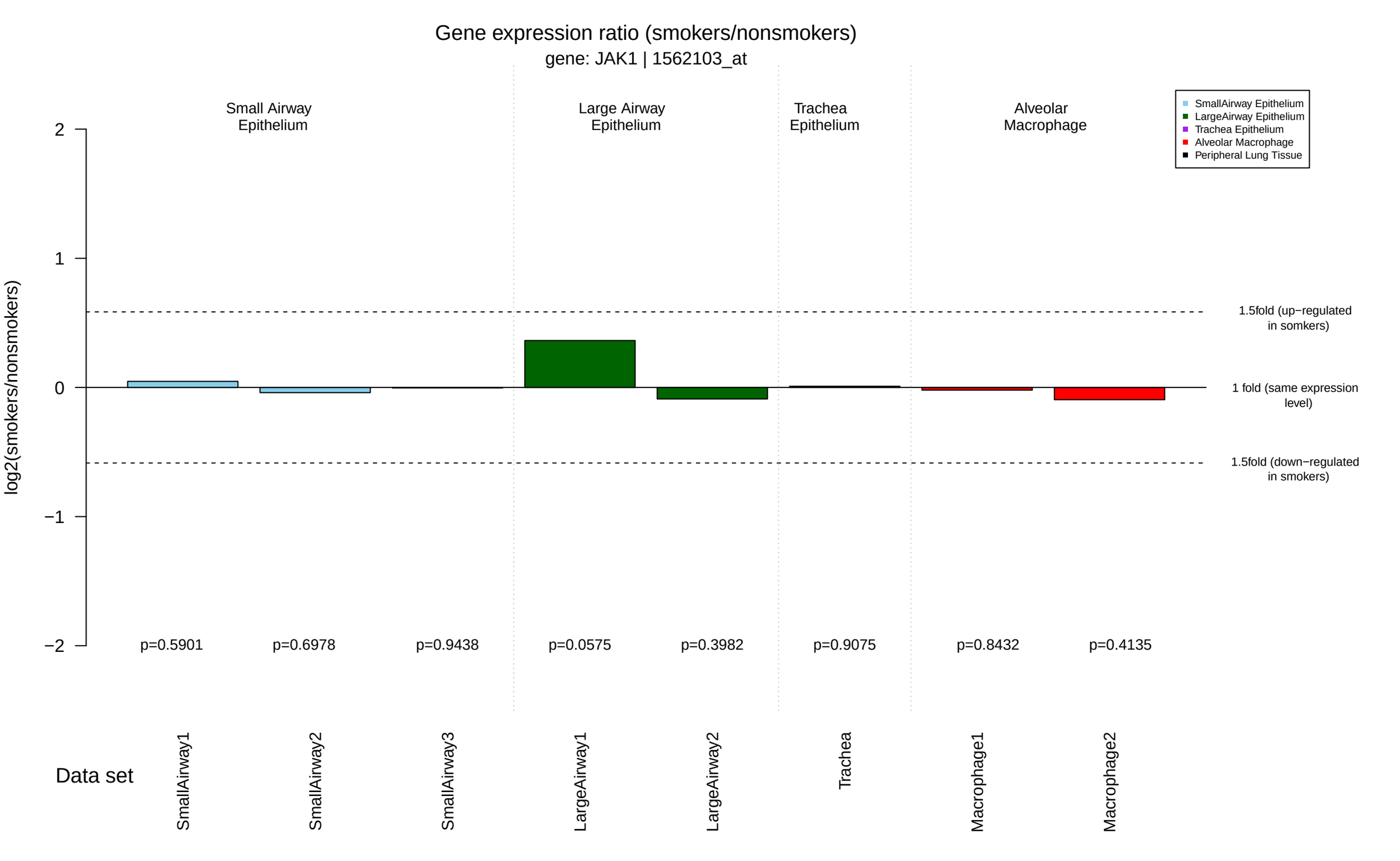
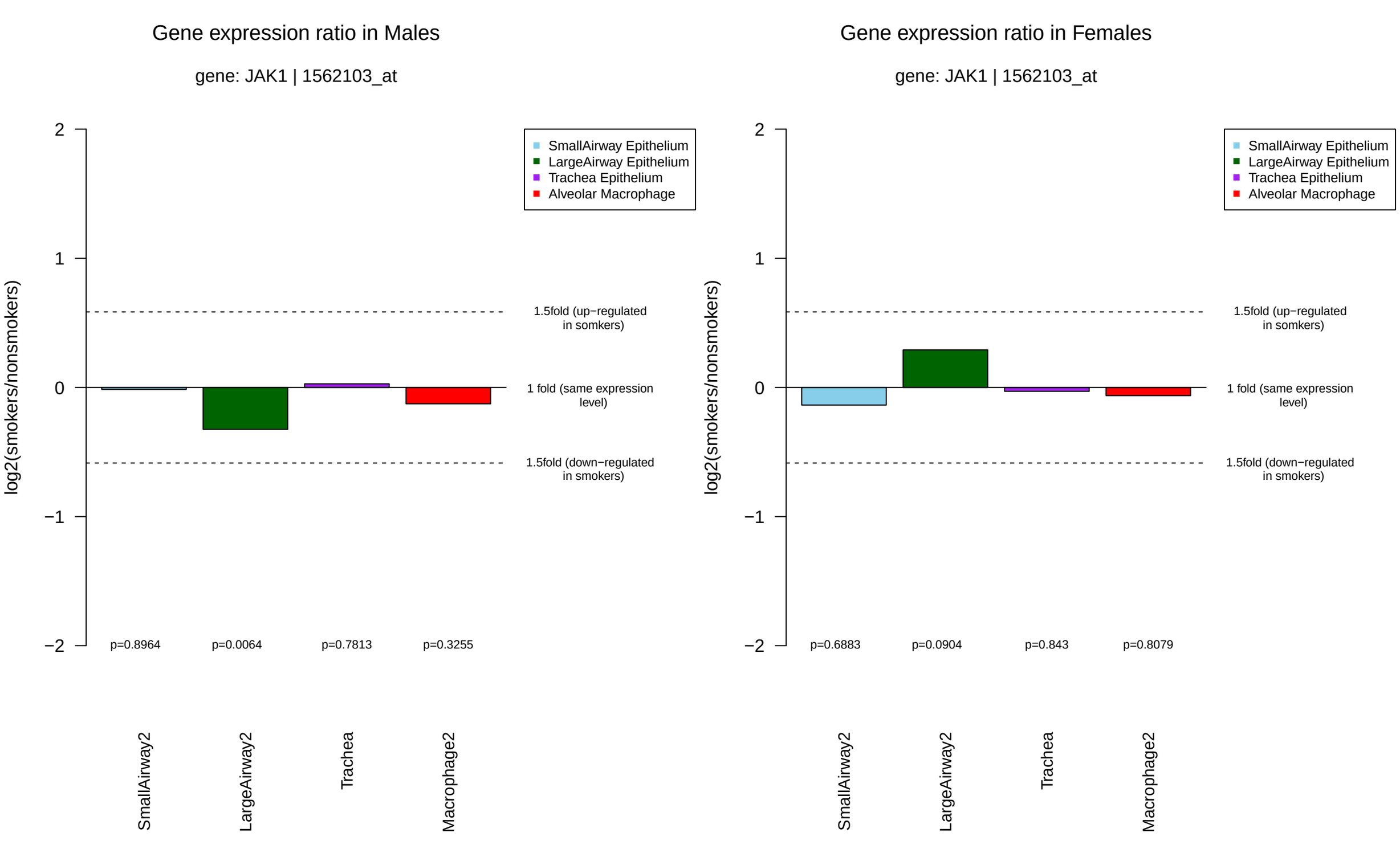
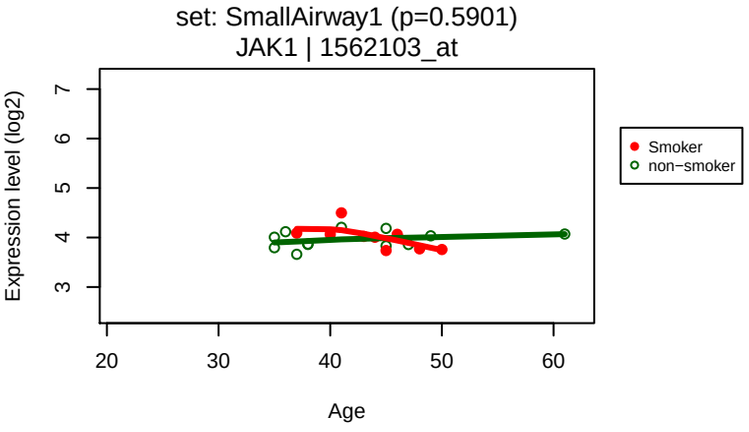
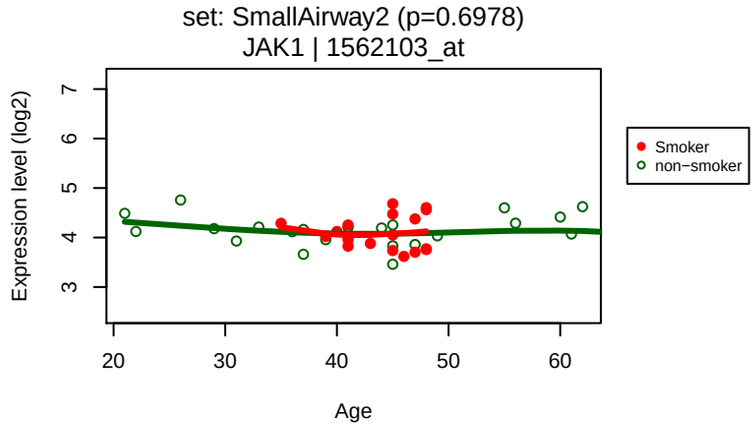
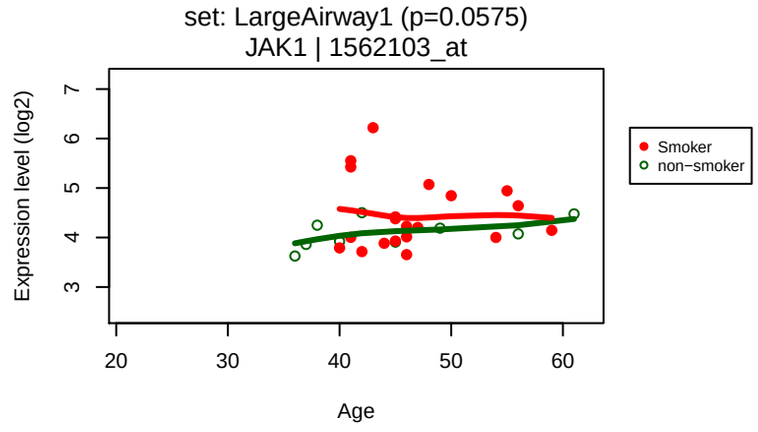

Gene expression of smokers and nonsmokers
gene: JAK1 | 1562103_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5901
p=0.6978
p=0.9438
p=0.0575
p=0.3982
p=0.9075
p=0.8432
p=0.4135
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

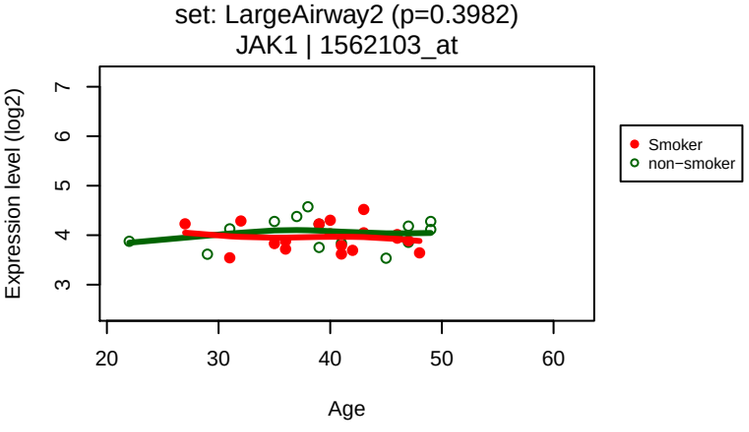

Gene expression ratio (smokers/nonsmokers)
gene: JAK1 | 1562103_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5901
p=0.6978
p=0.9438
p=0.0575
p=0.3982
p=0.9075
p=0.8432
p=0.4135
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

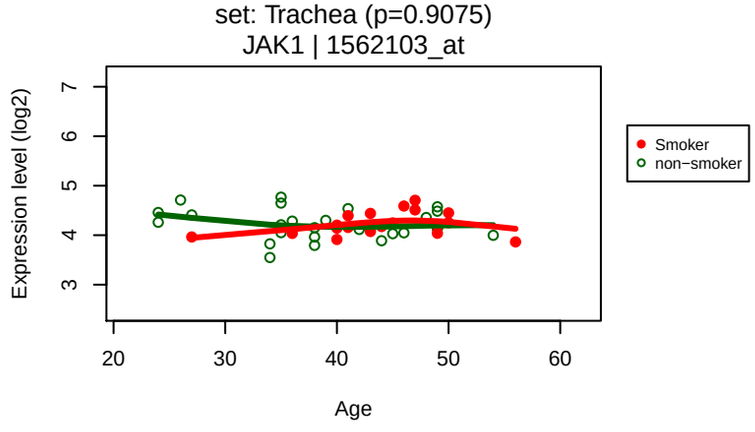

Gene expression ratio in Males
gene: JAK1 | 1562103_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8964
p=0.0064
p=0.7813
p=0.3255
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: JAK1 | 1562103_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6883
p=0.0904
p=0.843
p=0.8079
SmallAirway2
LargeAirway2
Trachea
Macrophage2

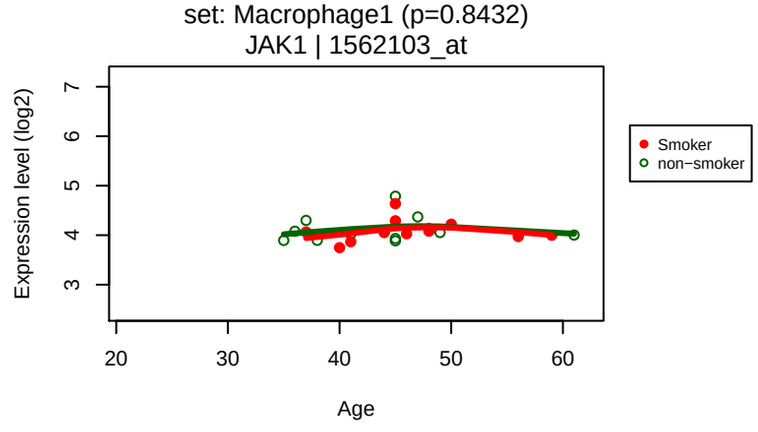

set: SmallAirway1 (p=0.5901)
JAK1 | 1562103_at
Expression level (log2)
Age
Smoker
non−smoker

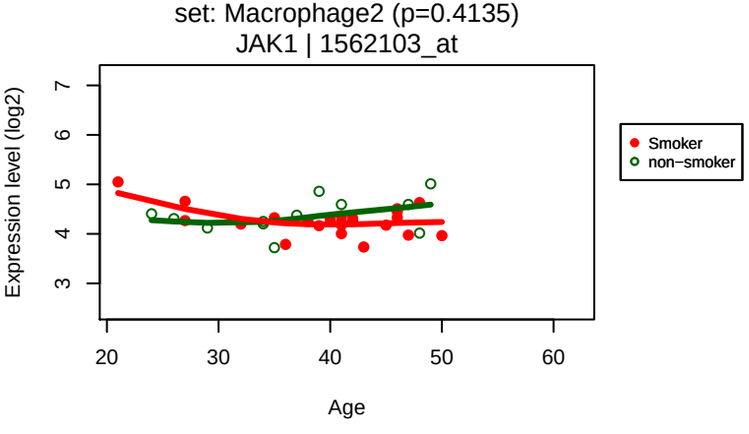

set: SmallAirway2 (p=0.6978)
JAK1 | 1562103_at
Expression level (log2)
Age
Smoker
non−smoker

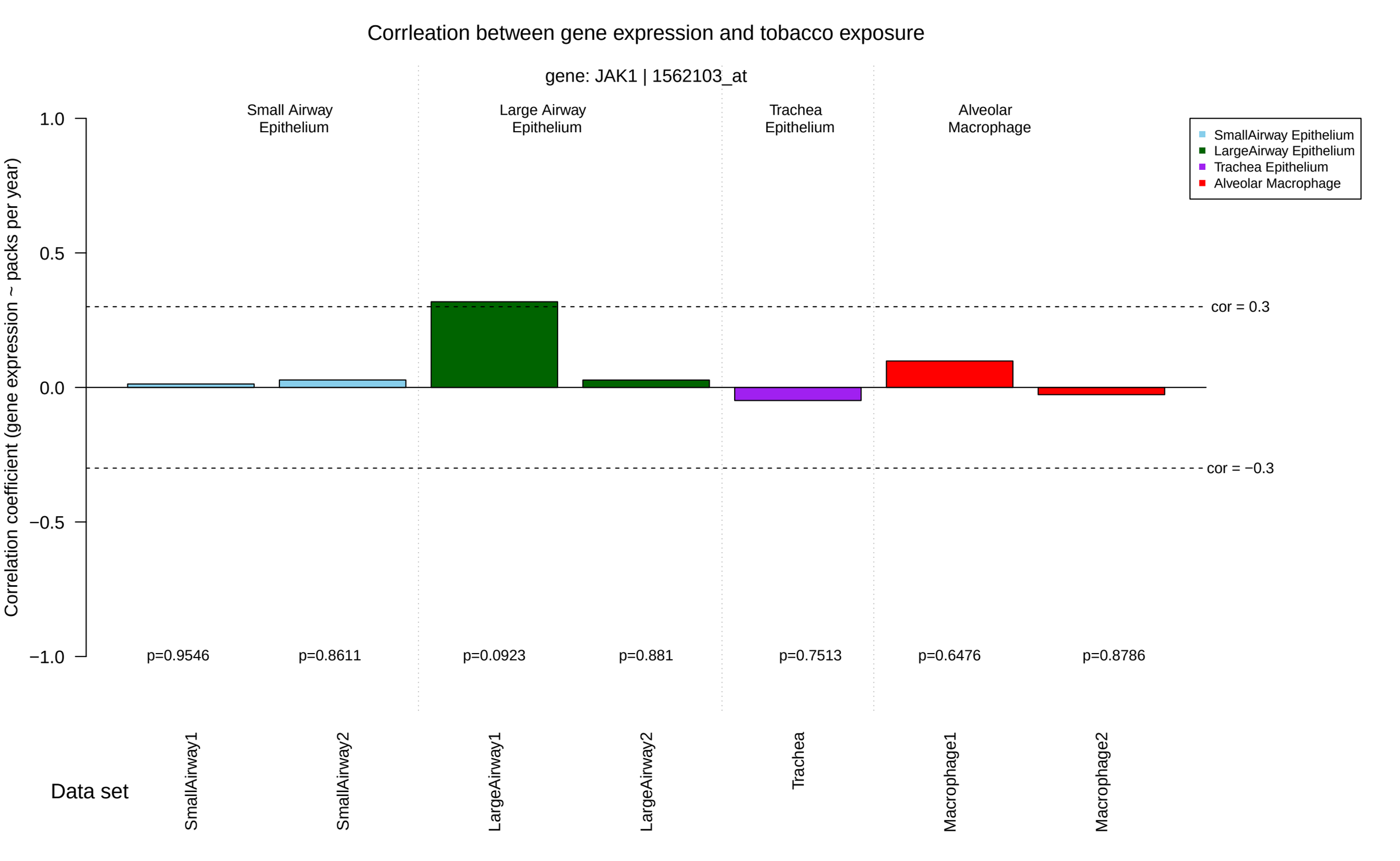

set: LargeAirway1 (p=0.0575)
JAK1 | 1562103_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3982)
JAK1 | 1562103_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9075)
JAK1 | 1562103_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8432)
JAK1 | 1562103_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4135)
JAK1 | 1562103_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: JAK1 | 1562103_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9546
p=0.8611
p=0.0923
p=0.881
p=0.7513
p=0.6476
p=0.8786
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2